Machine Learning for Mechanical Engineers at CoEP Session 8: Feature Selection

Session 8: FeatSel

Feature Extraction and Selection

From Raw Data to Useful Features

- Before training any model: which numbers do you feed it, and how many?
- **Feature Extraction**: raw data → a manageable set of numeric features.
- **Feature Selection**: given candidate features, which ones actually help?

Feature Extraction: Statistical Features

- Real signals often arrive as a stream: a sensor reading, a week of sales, a semester of test scores.
- A model needs one row per example, not 1000 raw readings, so summarize the stream instead.
- Simplest, most common summaries: mean, standard deviation, minimum, maximum, range.

Intuition

Describing a month of temperatures in one sentence? You wouldn't list 30 numbers, you'd say "averaged 22°C, swung between 18 and 27". That sentence *is* mean, min, max: statistical features.

Feature Extraction: A Worked Example

A sensor logs one reading a minute, for 10 minutes. Turn that into 5 summary features, one per sensor, per time window: several sensors, dozens of candidates.

```
import numpy as np

readings = np.array([21.1, 21.3, 21.0, 21.6, 22.0,
                     22.4, 22.1, 21.8, 21.5, 21.2])

features = {
    'mean':  readings.mean(),
    'std':   readings.std(),
    'min':   readings.min(),
    'max':   readings.max(),
    'range': readings.max() - readings.min(),
}
print(features)

# Output:
# {'mean': 21.6, 'std': 0.44, 'min': 21.0, 'max': 22.4, '
     range': 1.4}
```

Feature Selection: Which Features Actually Help?

Feature Selection: Ranking (Filter Methods)

- Score each feature *on its own* against the target, rank, keep the top few.
- "Score": a statistical test, e.g. an F-test (ANOVA) for a numeric feature vs. a class label.
- Fast, one pass, no model training. Called a **filter**: features filtered before any model sees them.
- Blind spot: scores features in isolation, can miss a feature that only helps *in combination*.

Ranking: Loading the Data

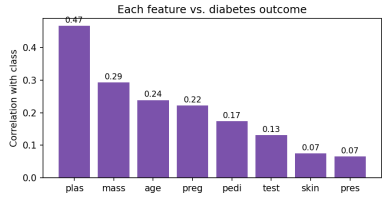
Pima Indians Diabetes dataset: 768 patients, 8 measurements, one outcome. `class`: 1 = diabetic, 0 = not. Which of the 8 columns actually predicts it?

```
import pandas as pd

url = "https://raw.githubusercontent.com/jbrownlee/
     Datasets/master/pima-indians-diabetes.data.csv"
names = ['preg', 'plas', 'pres', 'skin', 'test', 'mass', '
     pedi', 'age', 'class']
df = pd.read_csv(url, names=names)
df.head(3)

# Output:
#    preg  plas  pres  skin  test  mass   pedi  age  class
# 0     6   148    72    35     0  33.6  0.627   50      1
# 1     1    85    66    29     0  26.6  0.351   31      0
# 2     8   183    64     0     0  23.3  0.672   32      1
```

Ranking: Correlation with the Outcome

Simplest ranking: correlation of each column with `class`.

```
corr = df.corr()['class'].drop('class').sort_values(
     ascending=False)
print(corr.round(2))
```



Ranking: F-score (a Sharper Test)

Correlation is one lens; an F-test (ANOVA) is another, built for exactly this. Expect the same story as the correlation plot: `plas` dominating, `skin`/`pres` barely registering.

```
from sklearn.feature_selection import SelectKBest,
     f_classif

X, y = df.values[:, 0:8], df.values[:, 8]
skb = SelectKBest(score_func=f_classif, k=4).fit(X, y)
ranked = sorted(zip(names[:8], skb.scores_), key=lambda t:
      -t[1])

# F-score, highest to lowest:
# plas 213  mass 72  age 46  preg 40
# pedi  24  test 13  skin  4  pres  3
```

Feature Selection: Wrapper Methods

- Filter scores each feature alone. **Wrapper**: train/cross-validate the model on different feature *subsets*, let that score decide.
- Slower, many models trained, but catches combinations a filter would miss.
- Which subsets to try? Four classic strategies:
  - Exhaustive Search
  - Greedy Forward Selection
  - Greedy Backward Elimination
  - Best-First Search

Wrapper: Exhaustive Search

- Try *every* non-empty subset, keep the best cross-validated score.
- n features → $2^n - 1$ subsets. Pima's 8 → 255, small enough to actually run.
- Spoiler: dropping `skin` turns out to cost zero accuracy.

```python
import itertools
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import cross_val_score, KFold

model = LogisticRegression(max_iter=1000)
kfold = KFold(n_splits=5, shuffle=True, random_state=7)
best_score, best_subset = -1, None

for r in range(1, 9):
    for combo in itertools.combinations(range(8), r):
        s = cross_val_score(model, X[:, combo], y, cv=
     kfold).mean()
        if s > best_score:
            best_score, best_subset = s, combo

# Output (255 subsets evaluated):
# Best: all features except 'skin', accuracy=0.775
# All 8 features:                   accuracy=0.775
```

Wrapper: Why Exhaustive Search Doesn't Scale

- 8 features → 255 subsets: a few seconds.
- 3 sensors × 5 statistical features each = 15-20 candidates already.
- 20 features → $2^{20} - 1 \approx 1$ million subsets, roughly 4,000× more work.
- Realistic only for a small, already-narrowed feature set.

Wrapper: Greedy Forward Selection

- Start empty. Each round: add whichever remaining feature helps most.
- Stop when nothing left improves the score.

Same model and cross validation, on Pima:

Step	Feature added	Accuracy so far
1	plas	0.750
2	mass	0.767
3	pedi	0.773
4	pres	0.772

3 features reach 0.773, within 0.002 of exhaustive search's best, at a fraction of the cost.

Wrapper: Greedy Backward Elimination

- Start with *all* features. Each round: remove whichever hurts least (or helps).
- Stop once every remaining removal costs accuracy.

On Pima:

Step	Feature removed	Accuracy after
start	(all 8)	0.775
1	skin	0.775
2	age	0.772
3	preg	0.775

First move, drop `skin`, already matches what exhaustive search proved was right.

Wrapper: Best-First Search

- Greedy commits forever: once added or removed, never reconsidered.
- Best-first keeps a pool of promising subsets, not just one, and can expand any of them next.
- Can back out of an early choice greedy would be stuck with.
- Costs more than greedy, far less than exhaustive.

Intuition

Greedy: always take the steepest visible path uphill. Best-first: keep a mental map of your 3 best viewpoints, and choose which to climb from next.

Filter vs. Wrapper: Summary

Method	Looks at	Cost
Ranking (Filter)	One feature at a time	Cheapest
Exhaustive	Every possible subset	Guaranteed best, but explodes
Greedy Forward	Builds up, one add at a time	Cheap, no backtracking
Greedy Backward	Trims down, one remove at a time	Cheap, no backtracking
Best-First	A pool of candidate subsets	Middle ground

Intuition

No method is "correct." Ranking: fast first pass. Wrapper: costs more, sees combinations. Pick by budget.

Quick Check: Feature Selection

Think About It

The filter ranking on Pima placed `preg` (number of pregnancies) 4th out of 8 individually. Yet Greedy Backward Elimination's best 5-feature subset dropped `preg` entirely without hurting accuracy. Are these two results in conflict? Why or why not?

Quick Check: Feature Selection

Answer

- No conflict: they answer different questions.
- Filter score: how strongly `preg` relates to the outcome *by itself*, ignoring every other feature. On that measure alone it ranks 4th.
- Backward elimination asks a different question: given everything else already in the model, does `preg` add anything *on top of it*?
- `age` already correlates with number of pregnancies, so `preg`'s information is redundant: removing it costs nothing even though it looked useful alone.
- This is exactly the wrapper methods' advantage over filters: they can see redundancy that isolated ranking cannot.

Recap: Feature Extraction and Selection

- **Extraction**: raw data → manageable features (mean, std, min, max, range).
- **Selection**: which candidate features earn a place in the model.
- **Filter (Ranking)**: fast, scores features independently.
- **Wrapper** (Exhaustive, Greedy Forward/Backward, Best-First): slower, sees combinations filters miss.
- Narrows the search, you still decide what ships.

Hyperparameter Tuning

Hyperparameter Tuning

- **Parameters**: learned from data during `fit()`, e.g. regression coefficients, a tree's split points.
- **Hyperparameters**: set *before* training, control how learning happens, e.g. k in KNN, `max_depth` in a Decision Tree, C in an SVM.
- Picking by hand ("try $k = 3$, try $k = 5$, ...") doesn't scale past one hyperparameter.
- Fix: try many combinations, score each with cross validation, keep the best.

Intuition

Parameters: what the model learns. Hyperparameters: the knobs you set before it starts learning, like oven temperature and bake time, chosen before the cake goes in.

Hyperparameter Tuning: Grid Search

Try every combination on a fixed grid, keep the best cross-validated score.

```
import pandas as pd
from sklearn.model_selection import GridSearchCV
from sklearn.neighbors import KNeighborsClassifier

url = "https://raw.githubusercontent.com/jbrownlee/
     Datasets/master/pima-indians-diabetes.data.csv"
names = ['preg', 'plas', 'pres', 'skin', 'test', 'mass', '
     pedi', 'age', 'class']
df = pd.read_csv(url, names=names)
X, y = df.values[:, 0:8], df.values[:, 8]

model = KNeighborsClassifier()
param_grid = {'n_neighbors': [3, 5, 7, 9, 11, 15, 21]}

grid = GridSearchCV(model, param_grid, cv=5, scoring='
     accuracy')
grid.fit(X, y)

# Output: Best k 11, accuracy 0.749
```

Hyperparameter Tuning: Grid Search, Cost

- 7 candidate values of k $\times$ 5 folds = 35 model fits.
- All run automatically, no manual loop.
- One hyperparameter is cheap. Cost multiplies fast with more: 3 hyperparameters, 10 values each, = 1000 fits.

Hyperparameter Tuning: Randomized Search

- Grid Search checks every combination: fine for one hyperparameter, explodes with more.
- `RandomizedSearchCV`: sample a fixed number of random combinations instead.
- Trades a guarantee of the single best combination for a large cut in compute.

```
from sklearn.model_selection import RandomizedSearchCV

param_dist = {'n_neighbors': list(range(1, 30))}
random_search = RandomizedSearchCV(model, param_dist,
     n_iter=5,
                                   cv=5, scoring='
     accuracy', random_state=42)
random_search.fit(X, y)

# Output: tried 5 of 29 possible values -> Best k 13,
     accuracy 0.755
```

Hyperparameter Tuning: Randomized Search, Result

- Just 5 candidates out of 29 possible values of k.
- Matched Grid Search's result ($k = 11$ vs. $k = 13$, 0.749 vs. 0.755) at a fraction of the cost.
- In practice: often close to Grid Search's answer, rarely worse in any way that matters.

Quick Check: Hyperparameter Tuning

Think About It

Why is it important to use cross-validated scores, not a single train/test split, when comparing hyperparameter values during Grid or Randomized Search?

Quick Check: Hyperparameter Tuning

Answer

- A single split can hand a hyperparameter value a "lucky" or "unlucky" test set by chance.
- Averaging over k folds gives a far more reliable estimate of how each candidate value actually generalizes.
- So the value selected as "best" is best on average, not best on one arbitrary split.

Hands-On: End-to-End Mini Pipeline

The Dataset

- Steel Plates Faults dataset (UCI): a real manufacturing quality-control problem.
- 1941 steel plates, 27 geometric/luminosity measurements each, one of 7 recorded defect types per plate.
- Task: detect `K_Scatch`, a scratch-type surface defect, vs. everything else.
- 27 features: too many to exhaustively search, a real test for ranking and wrapper methods.
- Goal: pandas to load and inspect, sklearn to preprocess/select/train/evaluate, a revision lap before the algorithm-by-algorithm sessions start.

Step 1: Load and Look

```
import pandas as pd

feature_names = [
    'X_Minimum', 'X_Maximum', 'Y_Minimum', 'Y_Maximum', 'Pixels_Areas',
    'X_Perimeter', 'Y_Perimeter', 'Sum_of_Luminosity', 'Minimum_of_Luminosity',
    'Maximum_of_Luminosity', 'Length_of_Conveyer', 'TypeOfSteel_A300',
    'TypeOfSteel_A400', 'Steel_Plate_Thickness', 'Edges_Index', 'Empty_Index',
    'Square_Index', 'Outside_X_Index', 'Edges_X_Index', 'Edges_Y_Index',
    'Outside_Global_Index', 'LogOfAreas', 'Log_X_Index', 'Log_Y_Index',
    'Orientation_Index', 'Luminosity_Index', 'SigmoidOfAreas',
]
fault_names = ['Pastry', 'Z_Scratch', 'K_Scatch', 'Stains',
               'Dirtiness', 'Bumps', 'Other_Faults']

df = pd.read_csv('Faults.NNA', sep=r'\s+', header=None,
                 names=feature_names + fault_names)
df['target'] = df['K_Scatch']
print(df.shape)
df.head(3)

# Output: (1941, 35)
#    X_Minimum  X_Maximum  Y_Minimum  ...  target
# 0         42         50     270900  ...       0
# 1        645        651    2538079  ...       0
```

Step 2: Missing Values and Class Balance

```
print(df.isna().sum().sum(), "missing values")
print(df['target'].value_counts())

# Output:
# 0 missing values
# target
# 0    1550   (no K_Scatch defect)
# 1     391   (K_Scatch defect)
```

Step 3: Train/Test Split

Stratify on the target so both splits keep the same 80/20 class balance.

```
from sklearn.model_selection import train_test_split

X = df[feature_names].values
y = df['target'].values

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y)

# Output: X_train (1455, 27)   X_test (486, 27)
```

Step 4: Scale the Features

Fit the scaler on training data only, the same no-leakage rule from Data Preparation.

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler().fit(X_train)
X_train_s = scaler.transform(X_train)
X_test_s  = scaler.transform(X_test)
```

Step 5: Baseline, All 27 Features

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

model_all = LogisticRegression(max_iter=5000)
model_all.fit(X_train_s, y_train)
pred_all = model_all.predict(X_test_s)
print("Accuracy:", accuracy_score(y_test, pred_all))

# Output: Accuracy: 0.977
```

Step 6: Rank Features (Filter)

Same `SelectKBest` approach as before, now on 27 candidates instead of 8.

```
from sklearn.feature_selection import SelectKBest,
    f_classif

skb = SelectKBest(score_func=f_classif, k=10).fit(
    X_train_s, y_train)
top10 = [feature_names[i] for i in skb.get_support(indices
    =True)]
print(top10)

# Output: ['Pixels_Areas', 'X_Perimeter', '
    Sum_of_Luminosity',
#          'Minimum_of_Luminosity', 'Outside_X_Index', '
    Edges_Y_Index',
#          'LogOfAreas', 'Log_X_Index', 'Log_Y_Index', '
    SigmoidOfAreas']
```

Step 7: Retrain on the Top 10

```
idx = skb.get_support(indices=True)
model_10 = LogisticRegression(max_iter=5000)
model_10.fit(X_train_s[:, idx], y_train)
pred_10 = model_10.predict(X_test_s[:, idx])
print("Accuracy:", accuracy_score(y_test, pred_10))

# Output: Accuracy: 0.951
```

Step 8: Filter Result: Not a Free Lunch

- 27 features: 0.977 test accuracy.
- Filter top 10: 0.951 test accuracy.
- 17 dropped features weren't dead weight: accuracy fell by ∼2.6 points.
- Real trade-off: fewer features for a real accuracy cost. Worth it or not depends on the deployment.

Step 9: Try a Wrapper Instead

Greedy Forward Selection, cross-validated, on all 27 features:

```
from sklearn.model_selection import cross_val_score, KFold

Xs = StandardScaler().fit_transform(X)
kfold = KFold(n_splits=5, shuffle=True, random_state=7)
selected, remaining = [], list(range(27))

for step in range(6):
    scores = [(f, cross_val_score(LogisticRegression(
     max_iter=5000),
              Xs[:, selected + [f]], y, cv=kfold).mean())
      for f in remaining]
    best_f, best_s = max(scores, key=lambda t: t[1])
    selected.append(best_f); remaining.remove(best_f)

# Output: 6 rounds -> Log_X_Index, Steel_Plate_Thickness,
    Pixels_Areas,
#         Square_Index, X_Minimum, Orientation_Index
```

Step 10: Filter vs. Wrapper vs. Everything

Feature set	# Features	5-fold CV Accuracy
All features	27	0.975
Filter (top by F-score)	10	0.949
Wrapper (greedy forward)	6	0.963

Intuition

6 wrapper-chosen features close most of the gap to all 27 (0.963 vs. 0.975) while clearly beating the 10-feature filter (0.963 vs. 0.949): fewer, better-combined features can rival throwing everything at the model. The filter's top 10, chosen one at a time, missed a combination the wrapper found.

Step 11: Evaluate the Final Model

```
from sklearn.metrics import confusion_matrix,
    classification_report

wrapper_cols = ['Log_X_Index', 'Steel_Plate_Thickness', '
    Pixels_Areas',
                'Square_Index', 'X_Minimum', '
    Orientation_Index']
idx6 = [feature_names.index(f) for f in wrapper_cols]

model_6 = LogisticRegression(max_iter=5000)
model_6.fit(X_train_s[:, idx6], y_train)
pred_6 = model_6.predict(X_test_s[:, idx6])
print(confusion_matrix(y_test, pred_6))

# Output: [[384   4]
#          [ 16  82]]
```

Step 11: Classification Report

6 features, 96% test accuracy, beating both the 10-feature filter model and coming close to the full 27-feature model, on cross validation too, not just this one split.

```
print(classification_report(y_test, pred_6,
                            target_names=['no_k_scatch',
    'k_scatch']))

# Output:
#               precision  recall  f1-score  support
#  no_k_scatch       0.96    0.99      0.97      388
#     k_scatch       0.95    0.84      0.89       98
```

Quick Check: The Pipeline

Think About It

Step 6 fits `SelectKBest` on `X_train_s` (the scaled training set), not the full scaled dataset. Step 9's greedy forward selection, however, scores each candidate using cross validation on *all* of `X` (via `Xs`, not `X_train_s`). Is Step 9 leaking test information in a way Step 6 avoids?

Quick Check: The Pipeline

Answer

- No, both are safe, but for different reasons.
- Step 6 never touches `X_test_s` at all.
- Step 9 uses every row of `X`, including future test rows, but cross validation holds out a fold at a time: no fold's held-out data influences its own score.
- Step 9's real safety net: `X_test_s` from Step 3 was never part of its cross validation either, so Step 11's evaluation is still a fair, unseen check on the wrapper-selected features.

Try It Yourself

- Change `k=10` to `k=5` in Step 6: does accuracy hold up with even fewer features?
- Run Greedy Backward Elimination instead of Forward on all 27 features: same 6 features at the end?
- Swap `LogisticRegression` for a different classifier: does the winning feature set change?
- Try a different target fault type (e.g. `Bumps`): does the same feature set still win? (Spoiler: no, worth discussing why.)

Recap: The Pipeline Pattern

- Load → Inspect → Split → Scale → Select → Train → Evaluate → Compare.
- Every step reused today: pandas for loading/inspecting, sklearn for the rest.
- This shape doesn't change once you swap in a real ML algorithm: only Step 5/7's model does.
- Next: the algorithms themselves, one at a time, starting with Linear Regression.

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